

Vital Physiological Factors for ML-Based Stress Detection

Vital Factor	System Origin	Why It Matters	How ML Uses It
Cortisol	Endocrine (HPA Axis)	Primary biochemical marker reflecting cumulative and chronic stress exposure	Measured via biosensors or inferred indirectly from HRV, sleep disruption, and circadian rhythm features; ML learns personal sensitivity coefficient (k)
Heart Rate (HR)	Cardiovascular	Acute stress elevates heart rate while chronic stress shifts resting baseline	Time-series analysis to detect spikes, sustained elevation, and baseline drift
Heart Rate Variability (HRV)	Autonomic Nervous System	Reduced HRV indicates sympathetic dominance and reduced stress resilience	Extraction of features such as RMSSD, SDNN, and LF/HF ratios for classifiers or LSTM models
Galvanic Skin Response (EDA)	Sympathetic Nervous System	Rapid sweat gland activation provides immediate indication of arousal or stress	Phasic and tonic component separation with peak detection for real-time stress events
Skin Temperature	Peripheral Thermoregulation	Stress-induced vasoconstriction leads to measurable drops in skin temperature	Deviation detection relative to individual baseline using anomaly detection models
Pupil Dilation	Autonomic Nervous System	Pupil diameter increases with fear, anxiety, and cognitive workload	Computer vision models track temporal pupil diameter changes
Blood Pressure	Cardiovascular	Chronic stress elevates baseline blood pressure; acute stress causes transient spikes	Pattern recognition for abnormal variability and stress-related surges
Respiratory Pattern	Respiratory System	Stress causes rapid, shallow, and irregular breathing patterns	Signal processing combined with ML for rate, depth, and rhythm analysis
Sleep Architecture	Central Nervous System & Circadian	Stress disrupts REM and deep sleep, impairing physiological recovery	Sleep-stage classification and long-term trend modeling
Body Movement	Musculoskeletal	Anxiety causes restlessness while burnout or depression causes reduced activity	Accelerometer and phone sensor data used for behavioral profiling
Voice Characteristics	Vocal System / Limbic Input	Stress alters pitch, tempo, jitter, and pause structure during speech	Audio feature extraction such as MFCCs and pitch variance for emotion classifiers
Facial Muscle Tension	Facial Muscles / Emotional Reflexes	Micro-expressions and muscle tension reveal suppressed or unconscious stress	Computer vision models detect subtle facial muscle activation patterns